Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/04/2021
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	01/01/2019	**End Date:**	12/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Prof. Carlos Fernandez-Pello is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA Principal Investigator (PI) is Prof. Osamu Fujita, Hokkaido University. Co-PI is Professor Van Carey.		
COI Name (Institution):	Carey, Van Ph.D. (University of California, Berkeley)		
Grant/Contract No.:	80NSSC19K0331		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE this is continuation of "Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space," grant NNX14AF01G with the same principal investigator, Prof. Carlos Fernandez-Pello. Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. The objective of the proposed research program is to continue the experimental study of the flammability of wire materials in space exploration atmospheres and associated computational/theoretical tools to aid interpretation of test results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in spacecraft allows us to accurately elucidate the effect of the environment parameters on the ignition and flame spread over combustible material, and through them their potential fire hazard. Particularly important is the determination of the Limiting Oxygen Concentration (LOC) on flame extinction under spacecraft environments. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Another important aspect of the research is the effect of melting and dripping of plastic insulation in normal gravity in comparison with microgravity. The results are relevant because dripping will not occur in microgravity and consequently could impact their burning and methodology to screen. The investigation and results have also benefits for terrestrial fire safety by providing further information about the flammability of materials under a variety of environments.
Task Progress:	Abstract: A program to study the flammability characteristics of electrical wire insulations that will complement the existing JAXA project entitled “Fundamental Research on International Standard of Fire Safety in Space” (FLARE) is being carried out at U.C. Berkeley (UCB) under NASA sponsorship. The final objective of the project is to provide an assessment of wire materials flammability in the conditions expected in “Space Exploration Atmospheres” (SEA), i.e., low velocity flow, external radiant heating, reduced ambient pressure, and elevated oxygen concentration and microgravity. The near-term objective of the research is to develop and conduct tests in a ground-based FLARE/UCB apparatus that reflects the environments that are expected in space-based facilities, and that supports the International Space Station (ISS)/Kibo experiments to be conducted under the FLARE project. The apparatus is used to study the flammability of electrical wires, specifically ignition, flame spread and dripping of wire insulation during wire burning. Tests have been conducted with laboratory type wires consisting of a conducting core and PE insulation. During this reporting period these tests were temporarily halted because access to the UCB laboratories were restricted due to the Covid pandemic. For this reason, during this period the experimental task was supported by a theoretical task aimed to predict the experimental observations using an Artificial Neural Network (ANN). A reason to use ANN to model the wire flammability problem is that the problem is very complex and difficult to model and using ANN provides a path to model and predict wire flammability in spacecraft environments. A potential approach is to use an ANN to further understand of how certain variables in the wire flammability problem relate to one another and affect dripping/flame spread. This information can be used to model/predict both flame spread rate and dripping, and to determine predictive equation of the problem as a whole. Research Progress: Modeling Flame Spread and Dripping over Insulated Wires using a Neural Network. The work conducted during this reporting period provides a coherent understanding of flame spread parametric trends and associated fire safety issues in electrical systems for space applications. This understanding was obtained through use of an artificial neural network (ANN) that was trained to predict the flame spread rate along “laboratory” wires of different sizes and compositions [copper, nichrome, iron, and stainless-steel tube cores and HDPE (high density polyethylene), LDPE (low density polyethylene), and ETFE (ethylene tetrafluoroethylene) insulation sheaths] and exposed to different ambient conditions (varying flows, pressure, oxygen concentration, orientation, and gravitational strength). For these predictions, a comprehensive database of 1200 data points was created by incorporating flame spread rate results from both in-house experiments (400 data points) as well external experiments from other sources (800 data points). The predictions from the ANN showed that it is possible to merge together various data sets, including results from horizontal, inclined, vertical, and microgravity experiments, and obtain unified predictive results. While these initial results are very encouraging with an overall average error rate of 14%, they also show that future improvements to the ANN could still be made to increase prediction accuracy. Summary of Results. Training Data and Validation: The results from the ANN’s training showed a very strong agreement between the ANN’s predictions and the experimentally measured values, with an average error of 14%. However, it is expected to see good agreement here, as this is the data that was used to train the ANN. The true indication of the ANN’s performance is revealed by the validation results, which provide the results of the ANN’s validation with datasets divided by reference origin. Again, strong agreement was observed between the ANN’s predictions and the experimentally measured values, with an average error of 16% for data the ANN had never encountered before. Such strong agreement gives the indication that the selected input parameters of core and insulation densities, thermal conductivities, specific heats, and cross-sectional areas, oxygen concentration, pressure, flow velocity, and gravitational strength are all highly important to the wire flame spread problem. However, these results also show that there is great room for improvement in the ANN predictions. Parametric Trends: While there is room for improvement in the ANN model, that is not to say that nothing further can be learned from it in its current state. Examining some of the parametric trends predicted by the ANN can give further insight into the flame spread rate along electrical wire insulation problem. Because of the previous focus on spacecraft applications as well as the current focus on terrestrial transportation and structural applications, several parametric trends were examined. For the more spacecraft-related applications as well as limited terrestrial applications a parametric surface was obtained which demonstrates the ANN’s predictions for flame spread rate’s dependence on oxygen concentration and ambient pressure, spanning both the oxygen concentration and pressure ranges of the gathered dataset. The surface clearly displays the well-known trend for flame spread rate to increase both with increasing oxygen concentration and increasing pressure. However, it also shows further nuances of the problem that have not previously been well explored, such as the response of flame spread rate to high oxygen concentrations at low pressures

	as well as its response to high pressures at low oxygen concentrations. The ANN's predictions show that these parameters somewhat balance each other out and that even at extremely low oxygen concentrations or pressures, a high value of the other parameter can still result in a moderately fast flame spread rate. Aiming for further relevance to terrestrial transportation and structural applications, another parametric surface was obtained that demonstrates the ANN's predictions for flame spread rate's dependence on wire core cross-sectional area and axial flow velocity, spanning both the wire core diameter and axial flow velocity ranges of the gathered dataset with all other variables held at the same previous constants and this time with 21% oxygen concentration and 101 kPa (14.6 psi) ambient pressure. This surface shows that there is much more variation in flame spread rate when varying core cross-sectional area and axial flow velocity, suggesting that these parameters may be highly important for assessing fire safety. Another interesting feature that is found in the surface is the steep drop off in flame spread rate when the cross-sections area of the core approaches ~10 mm2, with some slight variation depending on the axial flow velocity. This dramatic decrease in flame spread rate may illustrate a critical core diameter, where the wire core transitions from being a source conducting heat toward the flame to a sink drawing heat out of it. Interestingly, this effect does not seem to be nearly as present in the opposed flow (negative axial flow velocity) regime. Future Work: The initial results from the ANN have shown to be very promising, and it is believed that the ANN could be used for much more in-depth analyses in the future. To begin with, only a very few number of flame spread rate parametric surfaces were examined in this study. Analysis of all the other possible flame spread rate parametric surfaces could bring so much further insight to the flame spread rate along electrical wire problem and potentially reveal results or nuances that have previously gone undiscovered. A sensitivity analysis examining each of the parameters affecting the flame spread rate along electrical wire insulation problem could also be performed using the predictions of the ANN. Such an analysis could be an immensely useful tool for shedding further light onto which parameters in the problem are the most important and influential when in comes to flame spread rate and assessing fire safety. Finally, the ANN could be expanded to include further parameters. As mentioned previously, some variables such as electric fields or electric currents and external radiation were excluded from this analysis. However, adapting the comprehensive database and the ANN to account for these variables could make the ANN predictions even more comprehensive and relevant to the current applications as well as additional ones.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Articles in Peer-reviewed Journals	Gagnon L, Carey VP, Fernandez-Pello C. "Using an artificial neural network to predict flame spread across electrical wires." Journal of Energy Resources Technology (JERT). 2021 Sep;143(9):092305. https://doi.org/10.1115/1.4050816 , Sep-2021
Dissertations and Theses	Gagnon L. (Lauren Gagnon) "Analyzing and Predicting Effects of Approximated Space Exploration Atmospheres on Flame Spread Rate across Various Types of Electrical Wires and Dripping of Molten Insulation." Ph.D. Dissertation, University of California Berkeley. August 2021. , Aug-2021
Papers from Meeting Proceedings	Gagnon L, Fernandez-Pello C, Urban J, Carey V, Konno Y, Fujita O. "Analyzing and Predicting Concurrent Flame Spread over Electrical Wires in Reduced Ambient Pressures." 12th U.S. National Combustion Meeting, Virtual, May 24-26, 2021. Meeting paper. 12th U.S. National Combustion Meeting, Virtual, May 24-26, 2021. , May-2021